Digital Image Processing

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Submission deadline: 1405/03/31 [Image Analysis] HomeWork 5

- Students are permitted a maximum of **8 day delays** throughout the course and you are allowed to use up to 5 days of your allowed delay per homework.
- Once the allowable delay period has expired, a **20% deduction** from the total assignment score will be applied **for each additional day of delay**.
- If external resources, such as **AI tools, websites or other references**, are utilized, proper **citation and attribution** are mandatory.
- Any instance of **plagiarism or cheating** will result in a score of **0** for all parties involved. A second violation will be considered **academic dishonesty** for the entire course, with further disciplinary actions as per institutional policies.
- Please upload all your answers in a file named **HW5_[SID]_[FullName].zip** to the **CW** system.

Theoretical ([50])

Question 1 [10]

You are analyzing a 5×5 grayscale image matrix, U, which contains an edge transition. Your goal is to apply standard edge detection operators to analyze the properties of the center pixel located at row 3, column 3 (assume 1-based indexing, so the center pixel value is 8).

The image matrix U is defined as:

$$U = \begin{bmatrix} 2 & 2 & 4 & 8 & 8 \\ 2 & 2 & 4 & 8 & 8 \\ 2 & 2 & 8 & 8 & 8 \\ 2 & 2 & 8 & 8 & 8 \\ 2 & 2 & 8 & 8 & 8 \end{bmatrix}$$

To analyze the center pixel, isolate the 3×3 neighborhood surrounding it:

$$W = \begin{bmatrix} 2 & 4 & 8 \\ 2 & 8 & 8 \\ 2 & 8 & 8 \end{bmatrix}$$

A)

The Sobel operator estimates the gradient of the image using a pair of 3×3 masks.

Given the horizontal mask (H_1) and vertical mask (H_2):

$$H_1 = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}, \quad H_2 = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

1. Calculate the bidirectional gradients g_1 and g_2 for the center pixel using the inner product (correlation) of the neighborhood W and the masks H_1 and H_2.
2. Compute the overall gradient magnitude g using the formula:

$$g = \sqrt{g_1^2 + g_2^2}$$

B)

Second-order derivatives are advantageous for detecting edges when the transition region is wider. Apply the discrete 8-neighbor Laplace operator mask:

$$L = \begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

Calculate the Laplacian response $\nabla^2 f$ for the center pixel.

Question 2 [15]

A)

Let $f(x, y)$ be an image and $G_\sigma(x, y)$ be a 2D Gaussian smoothing kernel. Prove mathematically that taking the spatial derivative of the smoothed image is equivalent to convolving the original image with the derivative of the Gaussian kernel. That is, demonstrate why the following holds true:

$$\frac{\partial}{\partial x}(G_\sigma * f) = \frac{\partial G_\sigma}{\partial x} * f$$

Provide a rigorous justification using the properties of the continuous convolution integral.

B)

Given your proof in Part A, we see that we can embed the derivative operation directly into the smoothing kernel. Conceptually, why is it practically problematic to bypass the Gaussian kernel and compute the first-order derivative directly on the raw image $f(x, y)$?

C)

Describe the specific imaging conditions where applying the Laplacian (or the Laplacian of Gaussian, LoG) is more advantageous than first-order gradient operators.

Question 3 [10]

A)

The mismatch energy at a given displacement (p, q) is defined as:

$$\sigma_{\eta}^{2}(p,q) = \sum_{m}\sum_{n}[v(m,n) - u(m-p, n-q)]^{2}$$

By expanding this equation algebraically, explain mathematically why finding the minimum mismatch energy is equivalent to maximizing the cross-correlation $c_{vu}(p,q) = \sum\sum v(m,n)u(m-p,n-q)$.

B)

According to the Cauchy-Schwarz inequality, under what exact mathematical condition does the absolute maximum of this cross-correlation occur?

Question 4 [15]

A)

Fisher's Linear Discriminant Analysis (LDA) aims to project data onto a vector w to best separate classes. State the specific mathematical objective function (the ratio) that Fisher's LDA seeks to maximize and identify what the matrices B and S represent in this equation.

B)

Consider a 2-dimensional feature space with four data points: $x_1 = (1,1)$, $x_2 = (2,1)$, $x_3 = (4,3)$, and $x_4 = (5,4)$. You are applying the K-means algorithm with $K = 2$. Assume your initial cluster centers are chosen as $c_1 = (1,1)$ and $c_2 = (5,4)$. Using the squared Euclidean distance as your metric, show the mathematical steps for the first full iteration.

Practical ([50])

Question 1[20]

Solve the Shape Moment section in the provided Jupyter Notebook.

Question 2 [20]

Solve the Template Matching section in the provided Jupyter Notebook.

Question 3 [10]

Perform the required experiment and answer the Conclusion section in the Jupyter Notebook.